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PHYSICS

Name of Candidate	JAYANT KUMAWAT		
Medium Eng./Hindi	ENGLISH	Registration Number	46104254
Center	ONLINE	Date	20th July 2026

INDEX TABLE			INSTRUCTIONS	
	Maximum Marks	Marks Obtained	1. Do furnish the appropriate details in the answer sheet (viz. Name, Registration Number and Test Code). 2. Question No. 1 is COMPULSORY. Of the remaining four questions (Q2 to Q5), any THREE are to be attempted. Thus a candidate has to attempt FOUR questions in all, totalling 200 marks. 3. The number of marks carried by a question/part is indicated against it. 4. Answers must be written in the medium authorized in the Admission Certificate, which must be stated clearly on the cover of this Question-Cum-Answer (QCA) Booklet in the space provided. No marks will be given for answers written in medium other than the authorized one. 5. Word limit in questions, if specified, should be adhered to. 6. Any page or portion of the page left blank in the Question-Cum-Answer Booklet must be clearly struck off.	
1(a)	10			
1(b)	10			
1(c)	10			
1(d)	10			
1(e)	10			
2(a)	20			
2(b)	20			
2(c)	10			
4(a)	15			
4(b)	15			
4(c)	20			
5(a)	15			
5(b)	15			
5(c)	20			
6(a)	15			
6(b)	20			
6(c)	15			
	250			
Total Marks Obtained: Remarks:			Is student recommended for One-to-One mentoring? Recommended Strongly Recommended	

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EVALUATION INDICATORS

1. Contextual Competence
2. Content Competence
3. Language Competence
4. Introduction Competence
5. Structure - Presentation Competence
6. Conclusion Competence

Overall Macro Comments / feedback / suggestions on Answer Booklet:

1.

2.

3.

4.

5.

6.

All the Best

My Total Attempt = 160marks/250m.

Extremely slow writing
Speed! + Improper
Time mgmt. during test.

भौतिकी / PHYSICS

Mechanics of Particles

Time Allowed: Three Hours

Maximum Marks: 250

QUESTION PAPER SPECIFIC INSTRUCTIONS

Please read each of the following instructions carefully before attempting questions:

*There are **Six** questions printed in **ENGLISH**.*

*Candidate has to attempt **Five** questions in all.*

*Questions no. **1** is compulsory and out of the remaining, any **four** are to be attempted.*

The number of marks carried by a question / part is indicated against it.

Answers must be written in the medium authorized in the Admission Certificate which must be stated clearly on the cover of this Question-cum-Answer (QCA) Booklet in the space provided. No marks will be given for answers written in a medium other than the authorized one.

Assume suitable data, if considered necessary, and indicate the same clearly.

Unless and otherwise indicated, symbols and notations carry their usual standard meanings.

Attempts of questions shall be counted in sequential order. Unless struck off, attempt of a question shall be counted even if attempted partly. Any page or portion of the page left blank in the Question-cum-Answer (QCA) Booklet must be clearly struck off.

Physical Constants:

Velocity of light	$c = 3 \times 10^8 \text{ m/s}$
Boltzmann constant	$k = 1.38 \times 10^{-23} \text{ J/K}$
Planck's constant	$h = 6.627 \times 10^{-34} \text{ J s}$
Permittivity of free space	$\epsilon_0 = 8.854 \times 10^{-12} \text{ F/m}$
Permeability of free space	$\mu_0 = 4\pi \times 10^{-7} \text{ H/m}$
Charge of the electron	$e = 1.6 \times 10^{-19} \text{ C}$
Mass of the electron	$m_e = 9.1 \times 10^{-31} \text{ kg}$
Universal gas constant	$R = 8.314 \text{ J mol}^{-1} \text{ K}^{-1}$
Universal gravitational constant	$G = 6.67 \times 10^{-11} \text{ Nm}^2/\text{kg}^2$

- Q1. (a) Derive the expression for centripetal and Coriolis accelerations for a particle moving in a rotating reference frame. 10
- (b) A particle of mass m is projected from the surface of Earth with velocity v . Derive an expression for its total mechanical energy and find the minimum velocity required for escape. 10
- (c) Explain and derive the parallel and perpendicular axis theorems for moment of inertia, and apply them to a thin circular disc. 10
- (d) Derive the expression for terminal velocity of a small sphere falling through a viscous fluid. 10
- (e) Derive the expression for relativistic addition of velocities and discuss its non-relativistic limit. 10

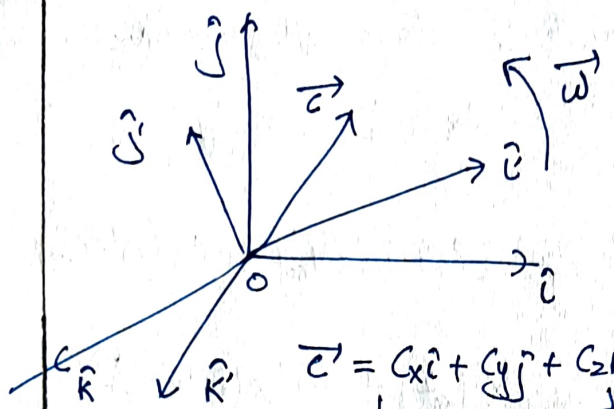
- Q2. (a) Derive the Lorentz transformations and show that they leave the spacetime interval invariant. 20
- (b) A uniform solid sphere rolls down an inclined plane without slipping. Derive its acceleration and time of descent. How does this compare with pure sliding motion? 20
- (c) Define gravitational self-energy and derive it for a uniform solid sphere. 10
- Q3. (a) Define expressions for the three elastic constants - Young's modulus, bulk modulus and shear modulus of an isotropic solid and establish the relations between them. Discuss their physical significance. 15
- (b) What do you understand by Laminar flow? Derive Poiseuille's equation for steady laminar flow of a viscous fluid through a horizontal capillary tube. 15
- (c) Explain the Michelson–Morley experiment, its null result, and its significance in the development of special relativity. 20
- Q4. (a) Discuss the concept of reduced mass and derive the two-body equation of motion in the centre-of-mass frame. 15
- (b) A gyroscope of mass m is spinning with angular speed ω and its axis is inclined at an angle θ to the vertical. Derive the expression for the precessional angular velocity. 15
- (c) Derive the Lorentz transformations and show that they leave the spacetime interval invariant. 20
- Q5. (a) Derive and explain Bernoulli's theorem for an incompressible fluid from the work-energy principle, and discuss its applications. 15
- (b) Explain the concept of four-momentum in relativity and derive the mass-energy relation. 15
- (c) Derive the equation of motion for a particle under a central inverse-square force. Discuss the nature of trajectories for bound and unbound motion. 20
- Q6. (a) Derive the relativistic energy–momentum relation
- $$E^2 = p^2 c^2 + m^2 c^4$$
- and discuss its significance for both massless and massive particles. Illustrate with the example of a photon and an electron. 15
- (b) Derive the expression for moment of inertia of a solid sphere and a hollow sphere about their diameters. Compare their rotational kinetic energies for the same angular velocity. 20
- (c) Derive Euler's equations for rotational motion of a rigid body and discuss their significance in describing torque-free motion. 15

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(a) Derive the expressions for centripetal acceleration and Coriolis acceleration for a particle moving in a rotating reference frame.

Q1(a)



Consider a frame S' rotating with $\vec{\omega}$ about origin O with respect to stationary frames S .

$$\vec{r} = C_x \hat{i} + C_y \hat{j} + C_z \hat{k} = \underbrace{C_x \hat{i} + C_y \hat{j} + C_z \hat{k}}_{\vec{r} \text{ as measured in } S} = \underbrace{C_x' \hat{i}' + C_y' \hat{j}' + C_z' \hat{k}'}_{\vec{r} \text{ as measured in } S'} \quad \text{--- ①}$$

Differentiating ① w.r.t time, we get

$$\frac{d\vec{r}}{dt} = \frac{dC_x}{dt} \hat{i} + \frac{dC_y}{dt} \hat{j} + \frac{dC_z}{dt} \hat{k} + C_x \frac{d\hat{i}}{dt} + C_y \frac{d\hat{j}}{dt} + C_z \frac{d\hat{k}}{dt}$$

For any vector transformation, $\frac{d\vec{B}'}{dt} = \vec{\omega} \times \vec{B}'$ --- ②

Using eqn ②, eqn ① transforms to,

$$\boxed{\frac{d\vec{r}}{dt} = \left(\frac{d\vec{r}}{dt} \right)_{\text{rotational frames } S'} + \vec{\omega} \times \vec{r}_{\text{rot}}} \quad \text{--- ③}$$

where $\vec{r}_{\text{rot}} = \text{Vector } \vec{r} \text{ as measured in } S'$.

Writing equation ③ for $\vec{r}$, we have,

$$\left(\frac{d\vec{r}}{dt} \right)_{\text{in}} = \left(\frac{d\vec{r}}{dt} \right)_{\text{rot}} + \vec{\omega} \times \vec{r}_{\text{rot}}$$

$$\therefore \vec{V}_{\text{in}} = \vec{V}_{\text{rot}} + \vec{\omega} \times \vec{r}_{\text{rot}} \quad \text{--- ④}$$

Now writing eqn ③ for $\vec{V}_{\text{in}}$, we have

$$\frac{d\vec{V}_{\text{in}}}{dt} = \left(\frac{d\vec{V}_{\text{in}}}{dt} \right)_{\text{rot}} + \vec{\omega} \times (\vec{V}_{\text{in}})_{\text{rot}}$$

Now substituting $\vec{V}_{in}$ from (4) in above eqn, we have

$$\vec{a}_{in} = \frac{d}{dt}(\vec{V}_{rot} + \vec{\omega} \times \vec{r}) + \vec{\omega} \times (\vec{V}_{rot} + \vec{\omega} \times \vec{r})$$

$$\vec{a}_{in} = \vec{a}_{rot} + \vec{\omega} \times \frac{d\vec{r}}{dt} + \frac{d\vec{\omega}}{dt} \times \vec{r} + \vec{\omega} \times \vec{V}_{rot} + \vec{\omega} \times (\vec{\omega} \times \vec{r})$$

$$\vec{a}_{in} = \vec{a}_{rot} + \frac{d\vec{\omega}}{dt} \times \vec{r} + 2(\vec{\omega} \times \vec{V}_{rot}) + \vec{\omega} \times (\vec{\omega} \times \vec{r})$$

$$\therefore \vec{a}_{rot} = \vec{a}_{in} + \underbrace{\left[-\frac{d\vec{\omega}}{dt} \times \vec{r} \right]}_{\text{Euler's acceleration}} + \underbrace{\left[-2(\vec{\omega} \times \vec{V}_{rot}) \right]}_{\text{Coriolis accel'n}} + \underbrace{\left[-\vec{\omega} \times (\vec{\omega} \times \vec{r}) \right]}_{\text{Centrifugal acceleration.}}$$

The forces accounting for Coriolis, centrifugal acceleration are fictitious in nature, that is, they are not a result of an actual physical interaction. Instead, they arise by observing the motion in rotating frame of reference (S').

1(b) A particle of mass m is projected from the surface of the Earth with velocity v . Derive an expression for its total mechanical energy and find the minimum velocity required for escape.

1(b)



The particle (m, v) experiences the gravitational force of attraction which is "central" in nature.

Consider the particle at r from the centre of earth at time ' t '.

Its total mechanical energy, $E_{\text{mech}} = K + U$.

$$E_m = \frac{1}{2}mv^2 - \frac{GMm}{r}$$

Since central forces are conservative and for conservative forces the mechanical energy remains conserved.

$$W = K_2 - K_1 \quad \text{and} \quad W = U_1 - U_2 \quad (\text{Conservative forces}).$$

$$\therefore U_1 + K_1 = K_2 + U_2 = E_{\text{mech}}.$$

For the particle to escape the gravitational field of earth, it must be projected with some v_{esc} .

Using work energy theorem, $W_{\text{net}} = \Delta K$

$$\therefore \int_R^{\infty} -\frac{GMm}{r^2} dr = K_{\infty} - K_R$$

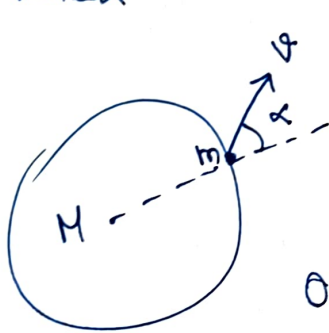
$$\text{At } \infty, K_{\infty} = 0 \quad \therefore \int_R^{\infty} \frac{GMm}{r^2} dr = \frac{1}{2}mv_{\text{esc}}^2.$$

$$GMm \left[-\frac{1}{r} \right]_R^{\infty} = \frac{1}{2}mv_{\text{esc}}^2$$

$$GM \left[-\frac{1}{\infty} + \frac{1}{R} \right] = \frac{1}{2}v_{\text{esc}}^2 \quad \therefore \boxed{v_{\text{esc}} = \sqrt{\frac{GM(2)}{R_e}} = \sqrt{2gR_e}}$$

$$v = \sqrt{2gR_e} = \sqrt{2 \times 9.8 \frac{m}{s^2} \times 6.4 \times 10^6 m} \approx 11.2 \text{ km/s.}$$

The results would have been same even if the particle was projected at an angle ' α ' with the vertical.



In plane polar coordinates,

$$\vec{v} = \dot{r}\hat{e}_r + r\dot{\theta}\hat{e}_\theta.$$

$$\therefore d\vec{r} = ()\hat{r} + ()\hat{\theta}$$

Only the $\hat{r}$ component of Gravitational force will do work.

1(c) Explain and derive the parallel axis theorem and perpendicular axis theorem for moment of inertia, and apply them to a thin circular disc.

(c) Parallel Axis Theorem - The moment of Inertia about any axis parallel to the axis passing through centre of mass (COM) of a rigid body can be expressed as

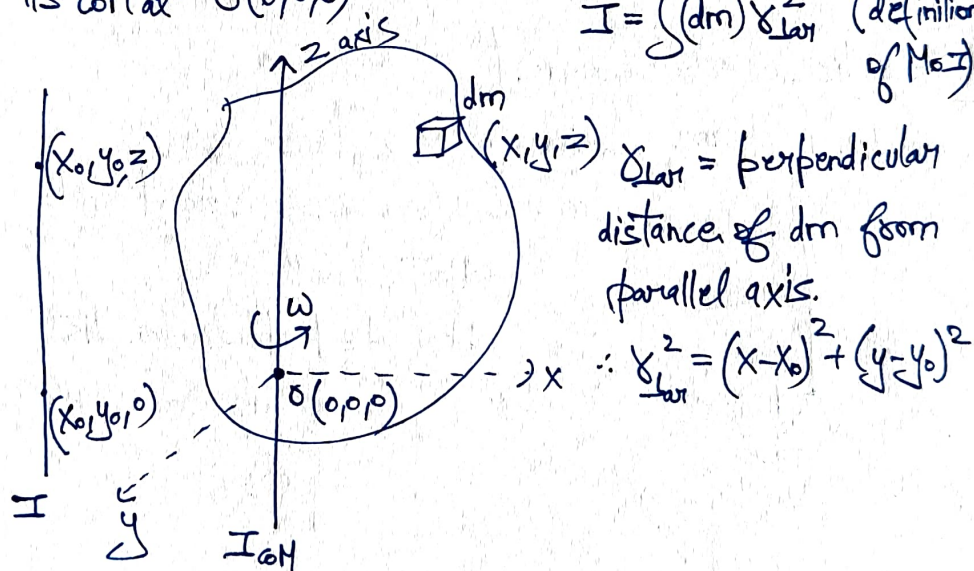
$$I = I_{\text{cm}} + Md^2$$

where d = perpendicular distance b/w the rigid body's COM axis & I .

$I_{\text{cm}} = MoI$ of rigid body about its COM.

Consider a rigid body rotating with $\vec{\omega} = \omega \hat{k}$, having its COM at $O(0,0,0)$.

$$I = \int (dm) r_{\text{cm}}^2 \quad (\text{definition of } MoI)$$



r_{cm} = perpendicular distance of dm from parallel axis.

$$\therefore r_{\text{cm}}^2 = (x - x_0)^2 + (y - y_0)^2$$

$$I = \int dm [(x - x_0)^2 + (y - y_0)^2]$$

$$I = \int dm (x_0^2 + y_0^2) + \int dm (x^2 + y^2) - 2x_0 \int dm x - 2y_0 \int dm y$$

Now writing the equation for coordinates of COM of rigid body,

$$x_{\text{com}} = \frac{\int dm x}{\int dm} = 0 \text{ (assumed COM at Origin)}$$

$$\therefore y_{\text{com}} = 0$$

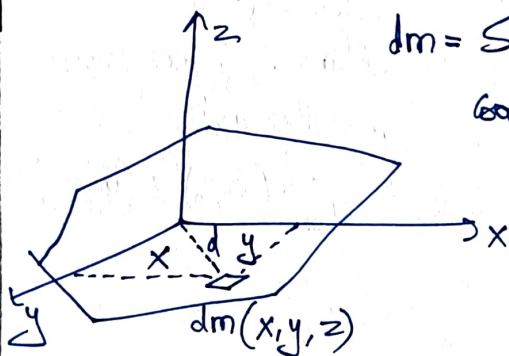
$$\therefore I = \int dm (x_0^2 + y_0^2) + \int dm (x^2 + y^2)$$

$$x_0^2 + y_0^2 = d^2 \therefore \boxed{I = Md^2 + I_{\text{com}}}$$

I_{com} in this case is about z axis, therefore $I_{\text{com}} = \int dm (x^2 + y^2)$

Perpendicular axis theorem

Consider a plane lamina in x, y plane.



dm = Surface mass element with coordinates x, y, z .

Now, d = perpendicular dist. of dm from z axis
 $= \sqrt{x^2 + y^2}$

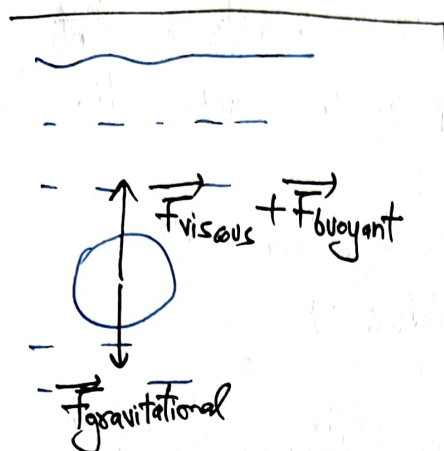
$$I_x = \int dm y^2, \quad I_y = \int dm x^2$$

Adding I_x & $I_y \Rightarrow I_x + I_y = \int dm (x^2 + y^2) = \int dm d^2$

$$\therefore \boxed{I_x + I_y = I_z} \text{ But valid ONLY for plane laminae objects.}$$

1(d) Derive the expression for the terminal velocity of a small sphere falling through a viscous fluid.

Q15 (d)



FBD of a falling sphere in a "homogeneous liquid medium"

STOKES LAW - A small & spherical body falling through a fluid experiences a backward dragging viscous force $= 6\pi\eta r v$, which results in absorption of energy from the medium in form of heat, where

η = dynamic viscosity of liquid.

v = speed of falling body.

When the viscous and buoyant force balances the gravitational force on sphere, it achieves a constant velocity called as "Terminal velocity".

$$\therefore \vec{F}_v + \vec{F}_b = \vec{F}_g \Rightarrow 6\pi\eta r v + \frac{4}{3}\pi r^3 \rho_f g = \frac{4}{3}\pi r^3 \rho g$$

where ρ = density of liquid

σ = density of sphere.

$$\therefore v = \frac{\frac{4}{3}\pi r^3 g (\sigma - \rho)}{6\pi\eta r} = \frac{2}{9} \frac{r^2 g}{\eta} (\sigma - \rho)$$

But this is only valid as per Stokes law if
 $\rightarrow$ The size of spherical body is small.

2) The medium is homogeneous & infinite in extent.
i.e. the intermolecular spacing in fluid $\ll$ diameter
of sphere.

3) The speed of body is less than critical
velocity suggested by Stokes $V_c = \frac{\eta}{\sigma \rho}$.

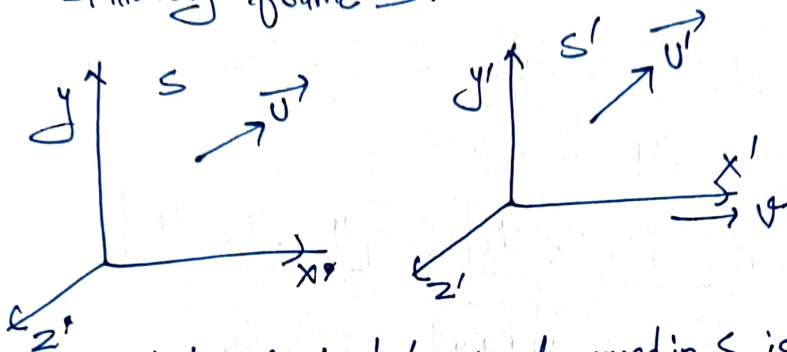
4) There are no waves or eddies setup in the
medium.

1(e) Derive the expression for relativistic addition of velocities and discuss its non-relativistic limit.

1(e)

Relativistic addition of velocities

Consider a frame S' moving with v in $+x$ direction w.r.t stationary frame S .



The velocity of particle as observed in S is $\vec{U} = U_x \hat{i} + U_y \hat{j} + U_z \hat{k}$.

From Lorentz transformation, the coordinates of any event can be transformed from S to S' as -

$$\begin{aligned} x' &= \gamma(x - vt) \\ t' &= \gamma\left(t - \frac{vx}{c^2}\right) \end{aligned} \quad \text{where } \gamma = \frac{1}{\sqrt{1 - \frac{v^2}{c^2}}} = \text{Lorentz factor of reference frame.}$$

$$\text{Now, } U'_x = \frac{dx'}{dt'} = \frac{d\left(\gamma(x - vt)\right)}{d\left(\gamma\left(t - \frac{vx}{c^2}\right)\right)} = \frac{U_x - v}{1 - \frac{vU_x}{c^2}} \quad \text{as } \frac{dx}{dt} = U_x$$

$$\therefore \boxed{U'_x = \frac{U_x - v}{1 - \frac{vU_x}{c^2}}}, \quad \boxed{U'_y = \frac{U_y}{\gamma\left(1 - \frac{vU_x}{c^2}\right)}}, \quad \boxed{U'_z = \frac{U_z}{\gamma\left(1 - \frac{vU_x}{c^2}\right)}}$$

The factor $\frac{1}{1 - \frac{vU_x}{c^2}}$ in U'_x ensures the distinction between

the classical addition of velocities & relativistic addition of velocities that speed of light is same in all inertial frames of reference and the Newtonian idea of absolute space and time doesn't fit well in relativistic regime.

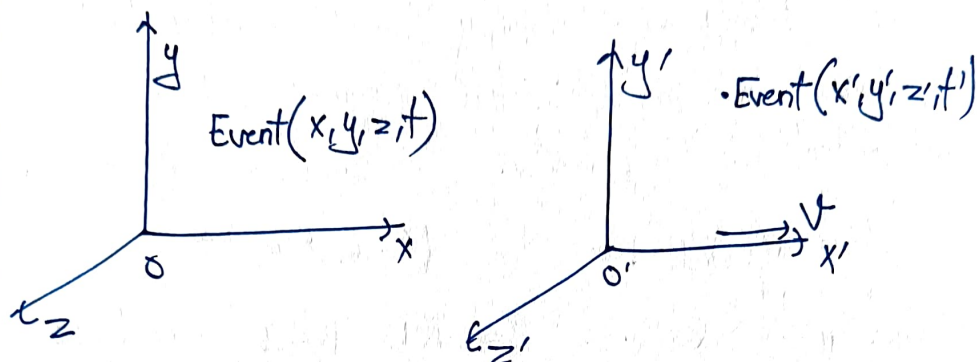
for $\frac{v}{c} \ll 1$ $\boxed{U_x' = U_x - v}$ or $\boxed{U_x = U_x' + v}$

Valid in classical mechanics
as all observers agree upon
the same time.

2(a) Derive the Lorentz transformations and show that they leave the spacetime interval invariant.

2.2(a)

Consider a frame S' moving with relativistic speed v along x direction w.r.t stationary frame S .



Lorentz transformation $\rightarrow$ transforms coordinates of any "Event" from S to S' or vice-versa.

$$\text{Let } \begin{cases} x' = Ax + Bt \\ t' = Cx + Dt \end{cases} \quad y' = y, \quad z' = z \quad \text{--- ①}$$

Event 1: Observe origin of S' in S .

$$\therefore S(vt, t) \text{ \& } S'(0, t')$$

$$\therefore 0 = Avt + Bt \text{ or } \boxed{B = -Av} \quad \text{--- ②}$$

Event 2: observe the origin of S in S' .

$$\therefore S(-vt', t') \text{ \& } S(0, t)$$

$$\begin{aligned} -vt' &= A(0) + Bt & \left| \quad t' &= C(0) + Dt \right| & \begin{aligned} t' &= Dt = At \\ \Rightarrow \quad \boxed{A = D} \end{aligned} \\ \therefore \quad \boxed{Bt = -vt'} & & \left| \quad \boxed{t' = Dt} \right| & & \end{aligned} \quad \text{--- ③}$$

Event 3: A light pulse emitted from O is observed along x \& x' axis.

$$S'(ct', t') \quad S(ct, t)$$

This is so because of the IInd postulate of Einstein's Special Theory of Relativity that speed of light remains same in all inertial frames and is independent of the motion of source or observer.

$$ct' = A(ct) + Bt \quad \& \quad t' = Cct + Dt$$

from eqⁿ ② & ③ i.e. $B = -AV$, $A = D$, we have

$$ct' = Act - AVt$$

$$t' = Cct + At$$

$$t' = A\left(t - \frac{Vt}{c}\right)$$

$$\therefore At\left(1 - \frac{V}{c}\right) = Cct + At$$

$$\boxed{t' = At\left(1 - \frac{V}{c}\right)}$$

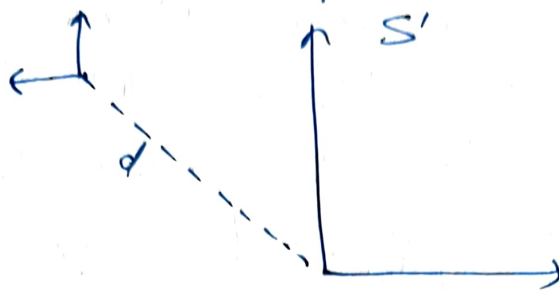
$$\therefore Cct = -\frac{AVt}{c}$$

$$\propto \boxed{C = -\frac{AV}{c^2}} \quad - \textcircled{4}$$

Event 4: Light pulse observed along y & y' .

$$\therefore S(0, ct)$$

But in S' , the light pulse has both the coordinates x' and y' , due to the relative motion between the frames.



Since $y' = y = ct$ and $x' = -vt'$.

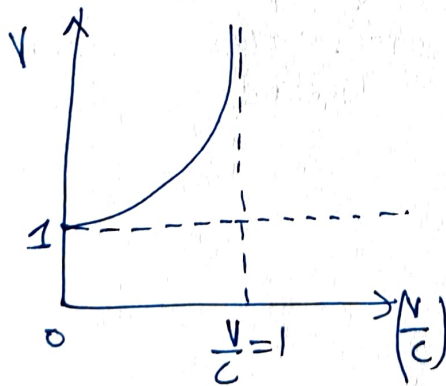
and $d = ct'$ (as the speed of light remains constant).

$$\therefore d^2 = x'^2 + y'^2$$

$$c^2 t'^2 = v^2 t'^2 + c^2 t^2$$

$$t'^2 = \frac{c^2 t^2}{c^2 - v^2} \quad \text{or} \quad \boxed{t' = \frac{t}{\sqrt{1 - \frac{v^2}{c^2}}}} = \gamma t$$

where $\gamma = \frac{1}{\sqrt{1 - \frac{v^2}{c^2}}} = \text{Lorentz Factor}$

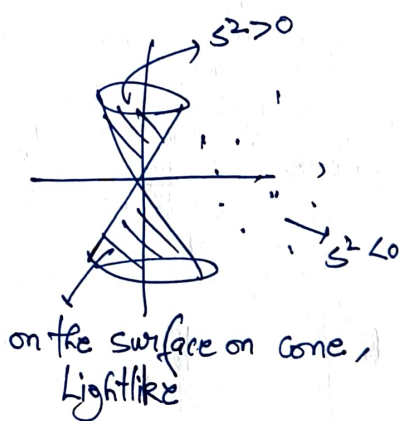


Invariance of Spacetime / Minkowski Interval
Treating (ct) as the 4th coordinate (Temporal),

The distance between two events in Minkowski Sp. is $s^2 = c^2 \Delta t^2 - \Delta x^2$ which remains invariant as we transform events from $S \rightarrow S'$ & vice-versa.

$$\begin{aligned}
 c^2(\Delta t')^2 - (\Delta x')^2 &= c^2 \left[\gamma \left(t - \frac{vx}{c^2} \right) \right]^2 - \gamma^2 (x - vt)^2 \\
 &= c^2 \gamma^2 \left[t^2 + \frac{v^2 x^2}{c^4} - \frac{2vxt}{c^2} \right] - \gamma^2 \left[x^2 + v^2 t^2 - 2xvt \right] \\
 &= t^2 \left[c^2 \gamma^2 - v^2 \gamma^2 \right] - x^2 \gamma^2 \left[1 - \frac{v^2}{c^2} \right] \\
 &= \underbrace{c^2 t^2 \gamma^2 \left[1 - \frac{v^2}{c^2} \right]}_1 - \underbrace{x^2 \gamma^2 \left[1 - \frac{v^2}{c^2} \right]}_1 = c^2 t'^2 - x'^2
 \end{aligned}$$

$$s^2 = \begin{cases} +ve & \text{Timelike} \\ -ve & \text{Spacelike} \\ 0 & \text{Lightlike} \end{cases}$$



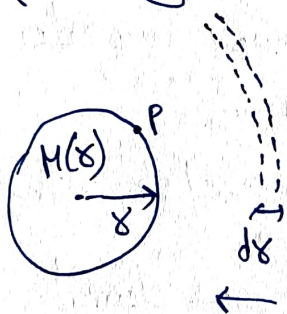
2(c) Define gravitational self-energy and derive it for a uniform solid sphere.

Q2(c)

Gravitational Self-energy - Work done in assembling the constituent mass elements ^{by} bringing them from infinity to their respective positions is the self energy of the body.

Consider a solid sphere of uniform mass density ρ and radius ' R '.

This solid sphere is assembled by integrating infinitesimally thin shells of thickness dx .



Work done in depositing a thin spherical shell on a solid sphere of radius ' x ' = $(dm) V(x)$,
 $V(x)$ = Potential of sphere at P which is only due to M_{enclosed} .

$$\therefore V(x) = -\frac{GM(x)}{x} = -$$

$$\& \quad dW = -\frac{GM(x)dm}{x} = -\frac{G}{x} \cdot (4\pi x^2 dx \rho) \cdot \frac{4\pi x^3 \rho}{3}$$

$$\therefore W = \int dW = -\int_0^R \frac{16G\pi^2 \rho^2}{3} x^4 dx$$

$$W = -\frac{16G\pi^2 \rho^2}{3} \cdot \left[\frac{x^5}{5} \right]_0^R = -\frac{16}{15} G\pi^2 \rho^2 R^5$$

$$g = \frac{M}{\frac{4}{3}\pi R^3} \quad \therefore g^2 = \frac{M^2}{\pi^2 R^6} \cdot \frac{9}{16}$$

$$\therefore R^5 g^2 = \frac{9}{16} \frac{M^2}{\pi^2 R}$$

$$\therefore W = -\frac{16}{15} \cdot G \pi^2 \left(\frac{9}{16} \frac{M^2}{\pi^2 R} \right)$$

$$W = -\frac{3}{5} \frac{M^2 G}{R}$$

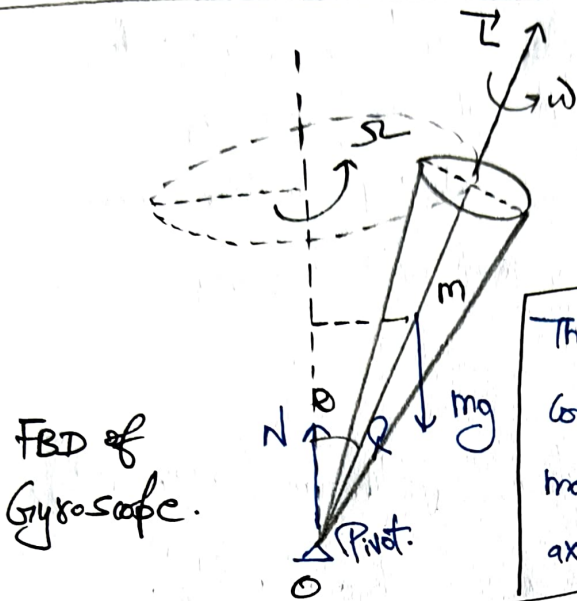
The negative sign indicates that energy was released while assembling the solid sphere.

$\frac{3M^2 G}{5R}$ is the amount of energy needed to supply,

so as to disperse its constituent mass elements back to infinity.

4(b) A gyroscope of mass m is spinning with angular speed ω , and its axis is inclined at an angle θ to the vertical. Derive the expression for the precessional angular velocity.

Q 4(b)



FBD of Gyroscope.

Let Ω = Precessional angular velocity of the Gyroscope.

The Gyroscope has two components of Angular momentum $\rightarrow$ Spin about its axis & orbital about the vertical axis.

Ignoring the small orbital angular momentum. The thing to note here

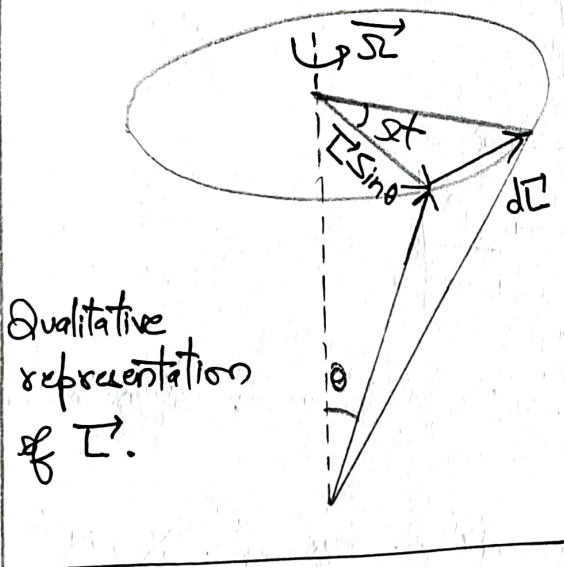
is "The direction of Spin Angular momentum vector is changing continuously".



There has to be an associated Torque causing $\vec{L}$ to change.

Assuming that, $L \sin \theta$ remains constant for a very short period ' dt ', we have

$$(L \sin \theta) \Omega dt = dL \Rightarrow \boxed{\frac{d\vec{L}}{dt} = \vec{\Omega} \times \vec{L}}$$



Qualitative representation of $\vec{L}$.

The pivot of Gyroscope doesn't constrain it to fall vertically downward but the Gyroscope precesses instead as the Normal reaction from pivot balances the weight of Gyroscope i.e.

$$\boxed{N = mg}$$

$\sum \vec{\tau}_O = 0$ for the vertical equilibrium (See FBD),
The torque of mg causes the change in L .

$$\therefore \boxed{\vec{\tau}_O = mgl \sin \theta}$$

$$\therefore mgl \sin \theta = \Omega L \sin \theta$$

$$\boxed{\Omega = \frac{mgl}{L} = \frac{mgl}{I_O \omega}}$$

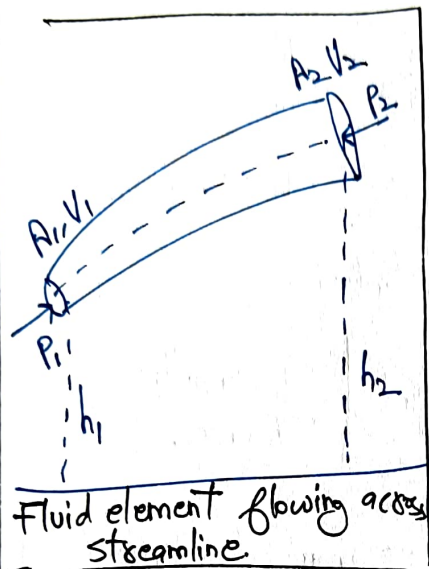
where Ω is independent of the angle of inclination ' θ '.

l = distance between COM of Gyroscope & pivot

5(a) Derive and explain Bernoulli's theorem for an incompressible fluid from the work-energy principle, and discuss its applications.

Q5)(a)

Bernoulli's theorem is a statement of conservation of Energy for fluids.



Consider a fluid element flowing along the streamline from A to B, considering the fluid as incompressible & inviscid. (no viscosity).

Total work done by the pressure = $P_1 A_1 V_1 \Delta t$
 W_1

But $A_1 V_1 \Delta t = \text{Volume of fluid element } (\Delta V)$

$$\& W_2 = -P_2 A_2 V_2 \Delta t \\ = -P_2 \Delta V$$

(Since $A_1 V_1 = A_2 V_2$ from the equation of continuity)

$$\boxed{W_{\text{net } P} = W_1 + W_2 = (P_1 - P_2) \Delta V} \quad \text{--- (1) continuity}$$

From work energy theorem,

$$W_{\text{net}} = \Delta K$$

$$\boxed{W_{\text{pressure}} + W_{\text{gravity}} = \Delta K}$$

$$W_{\text{gravity}} = -\Delta U = U_1 - U_2 \quad (\text{as } W = -\Delta U \text{ for conservative forces}). \\ = mg(h_1 - h_2)$$

$$W_{\text{grav.}} = \rho g (h_1 - h_2) \Delta V$$

$$(P_1 - P_2) \Delta V + \rho g (h_1 - h_2) \Delta V = \Delta V \left(\frac{\rho v_2^2}{2} - \frac{\rho v_1^2}{2} \right)$$

$$\therefore P_1 - P_2 + \rho g h_1 - \rho g h_2 = \frac{\rho v_2^2}{2} - \frac{\rho v_1^2}{2}$$

$$\therefore P_1 + \frac{1}{2} \rho v_1^2 + \rho g h_1 = P_2 + \frac{1}{2} \rho v_2^2 + \rho g h_2 = \text{Constant}$$

$$\therefore \boxed{P + \frac{1}{2} \rho v^2 + \rho g h = \text{Constant}} \quad \text{Bernoulli's Theorem.}$$

1) Static pressure $(P + \rho g h)$ + Dynamic pressure $(\frac{1}{2} \rho v^2)$ remains constant.

2) Pressure energy per unit volume + K.E per unit volume & potential energy per unit volume remain constant.

$$3) \underbrace{\left(\frac{P}{\rho g} \right)}_{\text{Pressure head}} + \underbrace{\frac{v^2}{2g}}_{\text{velocity head}} + h = \text{const.}$$

Applications

1) Venturimeter.

2) Designing the wings of airplanes.

3) The collapse of roofs during storm can be understood.

6(a) Derive the relativistic energy-momentum relation

$$E^2 = p^2 c^2 + m^2 c^4$$

and discuss its significance for both massless and massive particles.

Illustrate with the example of a photon and an electron.

Q5(a)

Derive the relativistic energy momentum relation,

$$E^2 = p^2 c^2 + m_0^2 c^4, \quad m_0 = \text{rest mass.}$$

Evaluating, $E^2 - p^2 c^2 = (\gamma m_0 c^2)^2 - (\gamma m_0 v)^2 c^2$

$$= \gamma^2 m_0^2 c^4 - \gamma^2 m_0^2 v^2 c^2$$

$$= \gamma^2 m_0^2 c^4 \left[1 - \frac{v^2}{c^2} \right]$$

$$E^2 - p^2 c^2 = m_0^2 c^4$$

$$\therefore \boxed{E^2 = p^2 c^2 + m_0^2 c^4}$$

For massless particles, $m_0 = 0$ (like photon etc.)

$\boxed{E = pc} \Rightarrow$ Even a massless particle can have momentum provided that it travels with the speed of light.

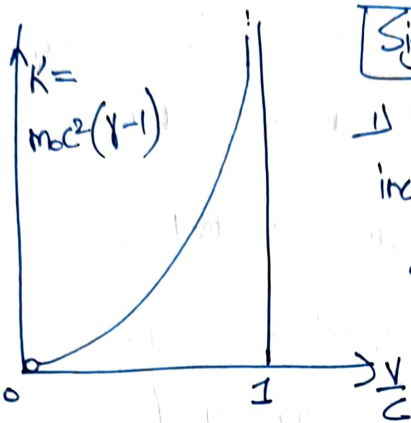
Now, $\boxed{K = m_0 c^2 (\gamma - 1)}$

$$E = m_0 c^2 + K \quad \therefore$$

$$K = m_0 c^2 \left[\frac{1}{\sqrt{1 - \frac{v^2}{c^2}}} - 1 \right] = m_0 c^2 \left[\left(1 - \frac{v^2}{c^2} \right)^{-\frac{1}{2}} - 1 \right]$$

For non-relativistic regime $\frac{v}{c} \ll 1 \therefore$

$$K = m_0 c^2 \left[1 + \frac{v^2}{2c^2} - 1 \right] = \left(\frac{m_0 v^2}{2} \right) = \text{Resemblance to classical counterpart.}$$



Significance

→ The kinetic energy increases without bounds as v approaches c which tells us that

"No massive particle can attain the speed of light".

For photons (massless particles).

$$E_i = E_f, \quad E = pc \text{ as the rest mass is } 0.$$

For electrons, the rest mass energy is 0.51 eV

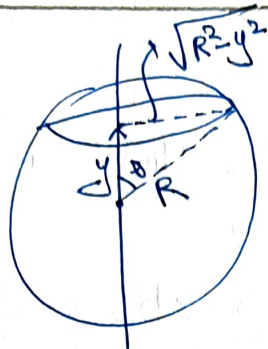
$$\& \text{ therefore in, } E_i = E_f = \sqrt{p^2 c^2 + m_0^2 c^4}$$

For massive particles, $K = \frac{1}{2} m_0 v^2$.

In relativity, the distinction between mechanical energy and other forms of energy disappears. Relativity treats all forms of energy on an equal footing.

6(b) Derive the expression for the moment of inertia of a solid sphere and a hollow sphere about their diameters. Compare their rotational kinetic energies for the same angular velocity

Q6(b)

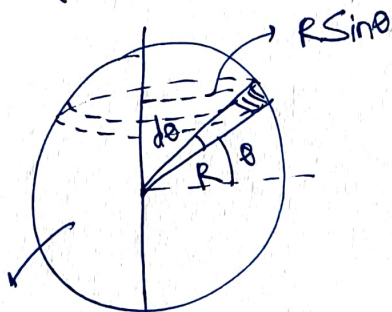


MoI of hollow sphere can be found by integrating all the rings.

$$dI = \int_{y=-R}^R \frac{dm \cdot (R^2 - y^2)}{2}$$

$$I = \int_{-R}^R \frac{(0) \cdot 2\pi \sqrt{R^2 - y^2} \cdot dy \cdot (R^2 - y^2)}{2}$$

& in (δ, θ)



MoI of every ring is $\int dm (R \sin \theta)^2$

where

$$dm = 2\pi (R \sin \theta) R d\theta \cdot \sigma$$

&

σ = Surface mass density of hollow sphere.

$$I = \int 2\pi (R \sin \theta)^3 R d\theta \sigma$$

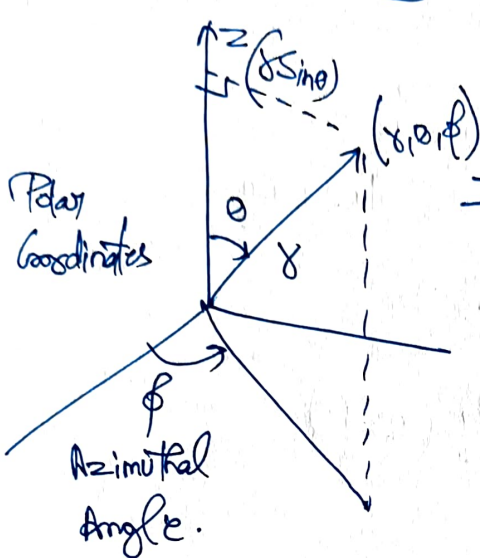
$$I = 2\pi R^4 \sigma \int_{-\pi/2}^{\pi/2} \sin^3 \theta d\theta$$

$$= 2\pi R^4 \sigma \cdot \pi \int_0^{\pi} \sin^3 \theta d\theta$$

$$= 2\pi R^4 \left(\frac{4}{3}\right) \cdot \frac{M}{4\pi R^2}$$

$$\boxed{I = \frac{2}{3} MR^2}$$

M.O.I of Solid sphere about z axis



$$I = \int (dm) (r \sin \theta)^2$$

$$I = \int (r d\theta \cdot r \sin \theta d\phi dr) (r \sin \theta)^2$$

$$I = \int_0^R r^4 dr \int_0^\pi \sin^3 \theta d\theta \int_0^{2\pi} d\phi$$

$$I = \int \frac{R^5}{5} \cdot \left(\frac{4}{3}\right) (2\pi)$$

$$= \frac{M}{\frac{4}{3}\pi R^3} \cdot \frac{R^5}{5} \cdot \left(\frac{4}{3}\right) \pi \cdot 2$$

$$\boxed{I = \frac{2}{5} MR^2}$$

6(c) Derive Euler's equations for rotational motion of a rigid body and discuss their significance in describing torque-free motion.

Q6) (c)

Euler's Equation of motion -

Choosing the principle axes of rotation for this derivation as "for a body of any mass distribution, it is always possible to find a set of 3 perpendicular axes about which the products of inertia vanish".

$$\vec{L} = \vec{I} \vec{\omega}$$

$$\vec{\omega} = \omega_1 \hat{i} + \omega_2 \hat{j} + \omega_3 \hat{k}, \quad \vec{I} = \begin{bmatrix} I_1 & 0 & 0 \\ 0 & I_2 & 0 \\ 0 & 0 & I_3 \end{bmatrix}$$

$$\therefore \vec{L} = I_1 \omega_1 \hat{i} + I_2 \omega_2 \hat{j} + I_3 \omega_3 \hat{k}$$

For any vector transformation,

$$\left(\frac{d\vec{L}}{dt} \right)_{\text{Space}} = \underbrace{\left(\frac{d\vec{L}}{dt} \right)_{\text{body}}}_{\text{Rate of change of spin angular momentum of the body}} + \vec{\omega} \times \vec{L}$$

Rate of change of spin angular momentum of the body.

$$\vec{L}_{\text{Space Coordinate System}} = \frac{d\vec{L}}{dt} + \vec{\omega} \times \vec{L}$$

$$\vec{\tau} = I_1 \dot{\omega}_1 \hat{i} + I_2 \dot{\omega}_2 \hat{j} + I_3 \dot{\omega}_3 \hat{k} + (\omega_1 \hat{i} + \omega_2 \hat{j} + \omega_3 \hat{k}) \times (I_1 \omega_1 \hat{i} + I_2 \omega_2 \hat{j} + I_3 \omega_3 \hat{k})$$

$$\vec{\omega} \times \vec{L} = \begin{vmatrix} \hat{i} & \hat{j} & \hat{k} \\ \omega_1 & \omega_2 & \omega_3 \\ I_1 \omega_1 & I_2 \omega_2 & I_3 \omega_3 \end{vmatrix} = \begin{pmatrix} (I_3 \omega_2 \omega_3 - I_2 \omega_2 \omega_3) \hat{i} \\ -\hat{j} (I_3 - I_1) \omega_1 \omega_3 \\ +\hat{k} (I_2 - I_1) \omega_1 \omega_2 \end{pmatrix}$$

$$\therefore \text{If } \vec{\tau} = 0 \Rightarrow$$

$$\begin{aligned} I_1 \dot{\omega}_1 &= (I_2 - I_3) \omega_2 \omega_3 \\ I_2 \dot{\omega}_2 &= (I_3 - I_1) \omega_1 \omega_3 \\ I_3 \dot{\omega}_3 &= (I_1 - I_2) \omega_1 \omega_2 \end{aligned}$$

Euler's dynamic eqⁿ of motion.